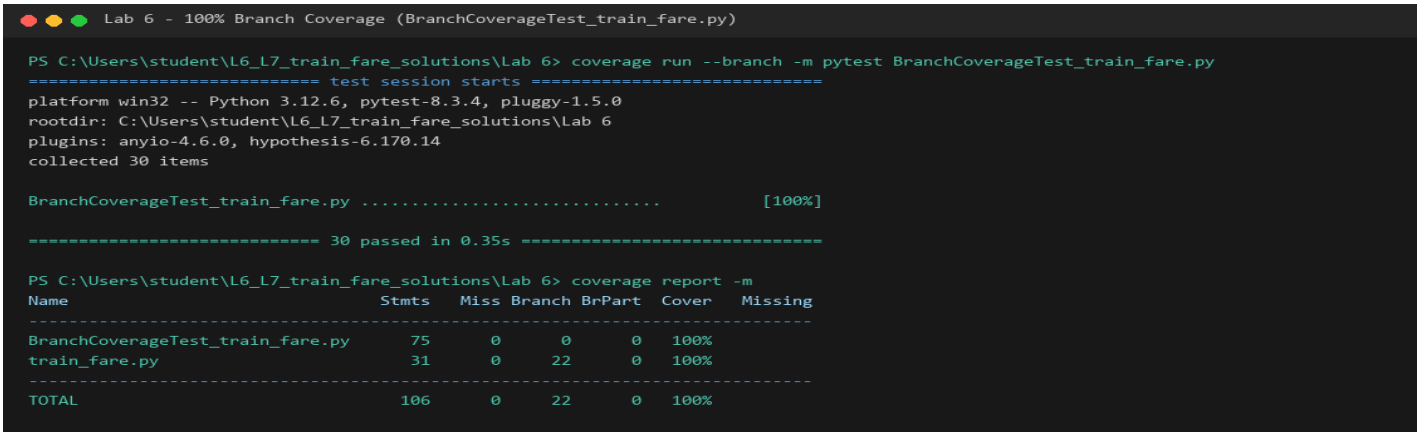
Train Fare Labs: Lab 6 & Lab 7 Comprehensive Report

Module: Software Engineering & Testing | **Target:** train_fare Application

Part 1: Lab 6 — Code and Branch Coverage Analysis

Objective: Measure and improve test coverage to achieve 100% statement and branch coverage. While statement coverage checks line execution, branch coverage verifies that every conditional path (if/elif/else) evaluates to True and False.

Test Suite	Files	Tests	Line Coverage	Branch Coverage
Initial Baseline	test_train_fare.py	2	45%	40%
Code Coverage Suite	CodeCoverageTest_train_fare.py	13	**100%**	100%
Branch Coverage Suite	BranchCoverageTest_train_fare.py	30	**100%**	**100%** (22/22)

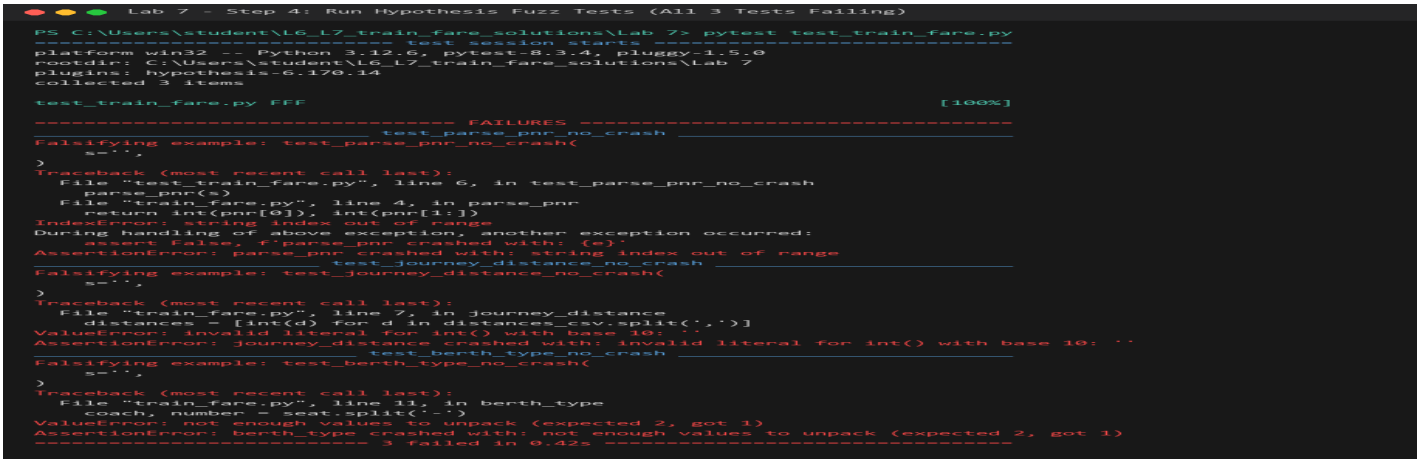


Part 2: Lab 7 — Fuzzing for Input Validation Bugs (Hypothesis)

Objective: Uncover edge-case crashes using property-based fuzz testing with Hypothesis (@given(st.text() | st.none())) and defensively harden the application code against malformed inputs.

- **parse_pnr:** Fuzzing found crashes on None (TypeError), empty string (IndexError), and non-digits (ValueError). Fixed by checking isinstance(str), stripping whitespace, and enforcing exactly 10 digits.
- **journey_distance:** Fuzzing found crashes on None (AttributeError), empty string / non-numeric tokens (ValueError), and single station inputs. Fixed by validating comma-separated integers and non-decreasing distance.
- **berth_type:** Fuzzing found crashes on None (AttributeError), missing hyphen (ValueError: not enough values to unpack), and invalid seat numbers. Fixed by enforcing coach-number format and positive seat numbers.

Hypothesis Failures on Buggy Code (Step 4):



All Fuzz Tests Passing on Hardened Code (Step 6):

